07 - Visualization

哈尔滨工业大学　黄子扬

2026 年 6 月 6 日

Contents

1 Anatomy of a ”Plot” / 绘图结构解剖

【Concept / 核心概念】1

Hierarchy / 层级结构 Matplotlib organize elements in a hierarchy of containers:

- **Figure**: The top-level container, representing the overall window or page. [1]
- **Axes / Subplots**: The actual plot area inside the figure where data is drawn. [1]
- **Axis**: Each Axes has an **XAxis** and a **YAxis**, which contain ticks, labels, and limits. [1]

简单来说：**Figure** 是画板，**Axes** 是画板上的坐标系，**Axis** 是坐标轴。

2 Figure Subplots Management / 画板与子图管理

2.1 Creation / 显式创建

【Example / 实操案例】1

Using plt.subplots() Explicitly create `figure` and `axes` objects to manage multiple plots. [4, 5]

```python
import matplotlib.pyplot as plt
import numpy as np

# Create a single plot
fig, ax = plt.subplots(nrows=1, ncols=1)

# Create a grid (2 rows, 4 columns)
fig, axes = plt.subplots(nrows=2, ncols=4, figsize=(10, 4))
```

2.2 Appearance Optimization / 视觉优化

【Warning / 避坑指南】1

Labels Overlapping / 标签重叠 If the figure is small, labels might overlap. Use `fig.tight_layout()` to automatically adjust margins. [4, 5]

3 The ”Language” of Matplotlib / 绘图视觉语言

3.1 Colors / 颜色规范 [2, 6-10]

- **Single letters**: ’r’ (red), ’b’ (blue), ’k’ (black), ’w’ (white).
- **Hex values**: e.g., `'0000FF'` for blue.
- **RGB[A] tuples**: Floats in range [11], e.g., `(1.0, 0.0, 0.0, 1.0)` for red.
- **Grayscale**: Strings from `'0.0'` (black) to `'1.0'` (white), e.g., `'0.75'`. [9, 12]

- **Cycle references**: Use `'C0'` to `'C9'` to reference the style's color cycle. [9, 12]

3.2 Markers Linestyles / 标记与线型 [10, 13, 14]

- **Markers**: `'.'` (point), `'o'` (circle), `'s'` (square), `'*'` (star). [10, 15]
- **Linestyles**: `'-'` (solid), `'--'` (dashed), `'-.'` (dash-dot), `':'` (dotted). [14, 15]

4 Core Plotting Functions / 核心绘图函数

4.1 Conditional Bar Plots / 条件条形图

【Example / 实操案例】1

Case Study: Negative Bars in Red [16]

```
x = np.arange(5)
y = np.random.randn(5)
fig, ax = plt.subplots()
vert_bars = ax.bar(x, y)
for bar, height in zip(vert_bars, y):
    if height < 0:
        bar.set(color='red') # Dynamic modification
```

4.2 Scatter for N-Dimensional Data / 多维散点图 [17, 18]

Scatter allows mapping data to x-position, y-position, **color (c)**, and **size (s)**.

```
sc = ax.scatter(x=x1, y=y, s=x2, c=x3, marker='o')
fig.colorbar(sc) # Add color scale bar
```

4.3 Image Display / 图像显示 [18]

Use `ax.imshow()` to display 2D data or images. Use `ax.axis("off")` to hide axis.

5 Advanced Decoration / 进阶装饰与细节

5.1 Mathtext (LaTeX) / 数学公式 [19, 20]

Surround text with `$` signs. Use `r` prefix for "raw" strings to avoid escape issues.

【Example / 实操案例】1

Formula Example `ax.set_title(r'`$\sigma\_i = \frac{3}{5}$`', fontsize=25)`

5.2 Legends Ticks / 图例与刻度 [3, 21, 22]

- **Legends**: Use `loc='best'` to automatically find the best location. [3, 23]
- **Ticks**: A **Tick** is the location of a **Tick Label**. [21]
- **Tick Params**: Use `ax.tick_params()` to change direction (e.g., `'in'`) or length. [22, 24]

5.3 Spines / 边框控制 [20]

Use `ax.spines['top'].set(visible=False)` to remove the top border of the plot.